

Matrix transforms in 2D

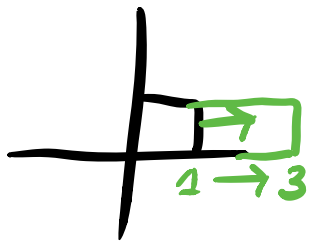
Basic skills

Q1. a) $\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \times 3 + 3 \times 1 \\ 3 \times 3 + 6 \times 1 \end{pmatrix} = \begin{pmatrix} 9 \\ 15 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 4 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$

c) $\begin{pmatrix} 5 & 0 & -5 \\ 6 & 7 & 3 \\ -4 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix} = \begin{pmatrix} -15 \\ -3 \\ -13 \end{pmatrix}$

Q2. a)



so

Points

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 3 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

so $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

b) by formula

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

c) Clockwise $270^\circ = \text{Anticlockwise } 90^\circ$
 so $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

d)  so $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$
 so $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} I_2$

e) by formula need $\tan \theta = 1$
 $y = (\tan \theta)x$ then $\theta = 45^\circ$ (or $\frac{\pi}{4}$)
 $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

$$\text{so } \begin{pmatrix} \cos(2 \times 45) & \sin(2 \times 45) \\ \sin(2 \times 45) & -\cos(2 \times 45) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Competency skills

a) $\begin{pmatrix} \cos 10^\circ & -\sin 10^\circ \\ \sin 10^\circ & \cos 10^\circ \end{pmatrix}$ by formula
 Anticlockwise rotation of 10°

b) $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = 4 I_2$ so enlargement of
 Scale factor 4


c) $\begin{pmatrix} 1 & 0 \\ 0 & 7 \end{pmatrix}$ stretch in y of scale factor 7



d) $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ stretch in x of scale factor 3



e) $\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$



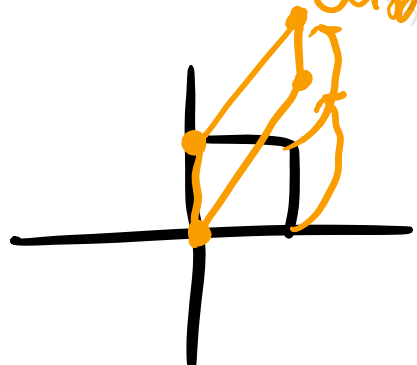
$$\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$$

Shear in x scale 6

Note. Some textbooks/sources may define slightly different

f) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$



$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

g) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$ rotation with $\cos \theta = \frac{\sqrt{3}}{2}$, $\sin \theta = \frac{1}{2}$
 so $\theta = 30^\circ$ so anticlockwise
 rotation of 30°

Q4. For $y = -2x$ use formula $y = (\tan \theta)x$

so $\tan \theta = -2 \Rightarrow \theta = -1.107$ ← radians

so
$$\begin{pmatrix} \cos(2x - 1.107) & \sin(2x - 1.107) \\ \sin(2x - 1.107) & -\cos(2x - 1.107) \end{pmatrix} = \begin{pmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{pmatrix}$$

Q5. As $M \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0$ ← zero matrix

it is not possible as $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ b \end{pmatrix}$

Q6. As $M \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, M \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix},$

$$M \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \end{pmatrix} \quad M \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

using $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ means

$$\begin{array}{llll} 2b = 0 & 5b = 0 & 3a + 5b = 3 & a + 3b = 1 \\ 2d = 2 & 5d = 5 & 3c + 5d = 8 & c + 2d = 3 \end{array}$$

so $b = 0, d = 1 \Rightarrow \begin{array}{ll} 3a = 3 & a = 1 \\ 3c = 3 & c = 1 \end{array}$

so $M = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

Q7. will need no change in y so

$$M = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \text{ for some } a, b$$

$$\text{Now } \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ 1 \end{pmatrix} \Rightarrow b = k$$

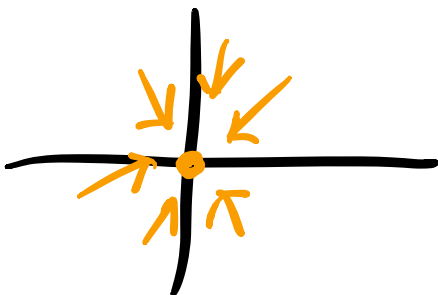
$$\text{and } \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow a = 1$$

$$\text{So } \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

Q8. The origin will always be the origin
as $M \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ so point must be origin

$$\text{and } M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ for all } x, y$$

$$\text{so } M = 0 \text{ zero matrix } \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$



Advanced skills

Q9. To map $\begin{pmatrix} x \\ y \end{pmatrix}$ onto $\begin{pmatrix} x \\ x \end{pmatrix}$ $\leftarrow y=x$
for any x

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ x \end{pmatrix} \Rightarrow \begin{aligned} ax + by &= x \\ cx + dy &= x \end{aligned}$$

so can let $a=c=1$, $b=d=0$

so $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ maps all to $y=x$

Q10. $NM = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix}$
 $\leftarrow$
order of application

$$= \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$
$$= \begin{pmatrix} \frac{5\sqrt{3}}{2} & -\frac{5}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

Q11. Points on line $y=2x$ are $\begin{pmatrix} x \\ 2x \end{pmatrix}$

$$\text{so } \begin{pmatrix} 4 & 8 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ 2x \end{pmatrix} = \begin{pmatrix} 20x \\ 3x \end{pmatrix}$$

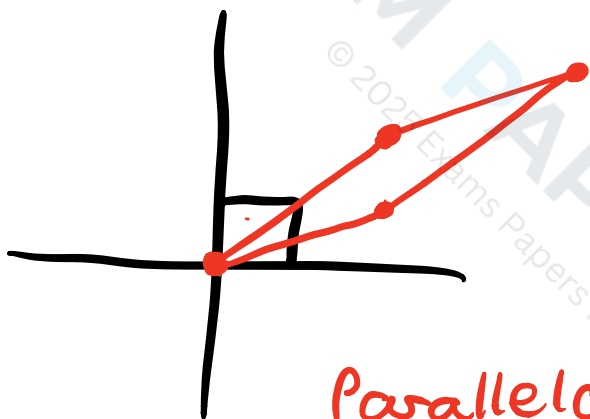
new coordinates $x' = 20x$, $y' = 30x$

$$\text{So } y' = \frac{3}{20}x'$$

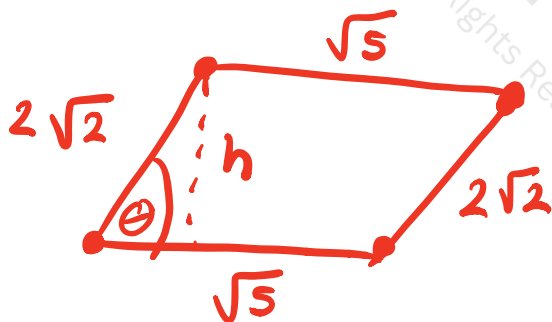
So $y = 2x$ is mapped to $y = \frac{3}{20}x$

Q12. $\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$

$$\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 4 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$



Parallelogram is base $\times$ perp height



Angle between $\underline{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ is

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} = \frac{6}{\sqrt{5} \times 2\sqrt{2}} = \frac{3}{\sqrt{10}}$$

For height $2\sqrt{2} \triangle h$ so $h = \sqrt{(2\sqrt{2})^2 - \left(\frac{6}{\sqrt{5}}\right)^2} = \sqrt{8 - \frac{36}{5}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$

$$a = 2\sqrt{2} \times \frac{3}{\sqrt{10}} = \frac{6}{\sqrt{5}}$$

So area is $\frac{2}{\sqrt{5}} \times \sqrt{5} = 2$ as required

Q13. need M s.t. $M \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\text{So } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow 3a + b = 1, \quad -4a + 2b = 0$$

$$3c + d = 0, \quad -4c + 2d = 1$$

$$\Rightarrow b = 1 - 3a \text{ so } -4a + 2(1 - 3a) = 0, \quad a = \frac{1}{5}, \quad b = \frac{2}{5}$$

$$d = -3c \text{ so } -4c + 2(-3c) = 1, \quad c = -\frac{1}{10}, \quad d = \frac{3}{10}$$

$$\text{Matrix is } \begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{1}{10} & \frac{3}{10} \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 2 & 4 \\ -1 & 3 \end{pmatrix}$$